

CareBeacon

Product Requirements Document

1. Overview

CareBeacon is a fall detection system for elderly users. It combines a wearable device with a web dashboard. The dashboard has two sides: an admin side for internal operators and a caregiver side for family members who monitor an elderly person.

2. Goals

- Detect a fall automatically and alert emergency contacts within seconds.
- Give caregivers a clear view of the monitored person's location, fall history, and device health.
- Give admin staff oversight across every registered family, device, incident, and support ticket.

3. User roles

Role	Description
Superadmin	Internal operator. Oversees all families, devices, incidents, and support activity. One admin tier only, so every sensitive action is logged in the audit log.
Caregiver	Registers on the site, links a device, and monitors one elderly person. A family can have more than one caregiver.
Monitored person	The elderly user who wears the device. Not a website account holder.

4. Registration and onboarding flow

1. A caregiver creates an account on the website.
2. The caregiver enters the device serial number to link it.
3. The backend checks the serial number against the provisioned device inventory (the Devices list on the admin side). An unrecognized serial number gets rejected, and the caregiver is told to check the number.
4. If the serial number has no existing family attached to it, the system creates a new family record and grants the caregiver access right away. The registration also enters an admin review queue. If admin finds a problem later, the family gets flagged or the account gets suspended.
5. If the serial number already belongs to an existing family, the system creates a pending join request instead of granting immediate access. Every existing caregiver on that family gets notified of the request.
6. An existing caregiver approves or denies the join request. Access to the family's location data, fall history, and device status only activates after approval. A denied or ignored request grants no access.

New family registrations grant access right away and get reviewed after the fact. Caregiver join requests on an existing family still block access until an existing caregiver approves.

5. Admin (superadmin) side

Analytics

Trends over time: daily and weekly active families, average incidents per week, device uptime. Growth charts for new signups and engagement charts for how often families use the app.

Families

Full list of every registered family account. Per family detail: monitored person, linked caregiver or caregivers, linked device or devices, account creation date, subscription status, and cumulative fall incident count. Search and filter by status, location, or device type. A "view as family" mode for troubleshooting, logged in the audit log. New registrations appear here for admin review. The caregiver already has access by the time it shows up, so review here is a check for problems rather than a gate.

Monitoring - Devices

Fleet wide list of every paired device: model, firmware version, battery, last sync, and online or offline status. Used to spot a device gone silent too long or a firmware version with known issues.

Monitoring - Incidents

Every fall or emergency event logged system wide. Detail per incident: what triggered it, when, which family, and the outcome (acknowledged, resolved, or false alarm).

Monitoring - Alert delivery

Tracks whether the notification for each incident actually reached the caregiver: SMS or push sent, delivered, or failed. This is the check that confirms the safety feature works end to end, not just that the sensor fired.

Support - Tickets

Inbound support requests from families, with status (open, in progress, resolved) and assignment.

Support - Audit log

Record of every sensitive admin action: who viewed which family's data, who used "view as family," who exported data, and when.

System - Settings

Notification and alert rule defaults, including inactivity thresholds and escalation timing. Data retention policy for location and incident history. Security settings: two factor authentication requirement and session timeout. Consent and privacy templates, plus terms of service versioning.

6. Caregiver side

- Live map showing the monitored person's current location.
- History of fall events with timestamp and confirmation status.
- Running fall incident count for the monitored person, going up by one each time an alert broadcasts. Cancelled alerts and SOS presses do not add to this number.
- Device status: battery level, signal strength, and whether the device is online, offline, or broken.
- Pending join requests from new caregivers trying to link the same device, with approve or deny actions.
- Ability to confirm an active alert directly from the dashboard, in addition to replying CONFIRM by SMS.

7. Alert and notification flow

Automatic fall detection

1. The device continuously reads motion data from the MPU6050 sensor.
2. The onboard algorithm or TinyML model flags a possible fall.
3. The device runs a local pre-alert: buzzer, LED, and vibration motor activate, and a 15 second cancel window opens.
4. If the user cancels within the window, the device aborts the alert and can send an SMS Cancel message to all contacts. The dashboard marks the event Cancelled, and the fall incident count does not increase.
5. If not cancelled, the device sends an SMS and places a voice call to every emergency contact at the same time, using the SIM7080G module directly. This path works independent of wifi and the app.
6. At the same time, the device sends the event data (fall event, timestamp, GPS coordinates, status UNCONFIRMED) to Firebase over wifi, or over cellular data if wifi is unavailable.
7. Once Firebase receives the event, the backend increments the fall incident count for that monitored person by one and sends a push notification to the caregiver dashboard or app. The running count is visible to every caregiver on the family and appears in that family's record on the admin side. The push notification requires Firebase Cloud Messaging set up on the frontend.
8. The dashboard shows the event as Unconfirmed with the monitored person's location.
9. The first confirmation from any channel, an SMS reply, an answered call, or a dashboard button, sets the status to Confirmed for every caregiver and stops all further automated alerts. This is the spam prevention behavior.

Manual SOS

1. The user presses the SOS button on the device.
2. The system skips motion detection and goes straight to the alert broadcast step above.
3. All emergency contacts get an SMS and a voice call immediately.
4. The dashboard shows the SOS event as Unconfirmed with the current location.

SOS events do not add to the fall incident count, since they are not falls.

Open item: the delivery method for caregiver approval requests (push notification, SMS, or in app only) is not yet decided.

8. System architecture

Device hardware

- ESP32-C3 as the main controller.
- MPU6050 for motion and fall detection.
- u-blox MAX-M10S for GPS location.
- SIM7080G for SMS, voice calls, and cellular data fallback.
- MEMS microphone, PAM8302 amplifier, and a mini speaker for two way voice.
- TP4056 charging module and MAX17048 fuel gauge for power management.
- SMD LEDs, a buzzer, and a coin vibration motor for local feedback.

Backend and frontend

- Firebase for the real time database and hosting, with PHP and MySQL as an optional alternative.
- HTML, CSS, and JavaScript for the web dashboard.

9. Design references

- Light theme with a muted blue primary color (#2F5480), reserved for buttons, navigation, and active states.
- Status colors carry all status meaning: green for good or online, amber for warning or low, red for alert or critical.
- A superadmin dashboard mockup already exists, covering the sidebar navigation, stat cards, family table, and incident feed described in this document.

10. Non functional requirements

- Two factor authentication available as a security setting.
- Configurable session timeout.
- Configurable data retention period for location and incident history.
- Every sensitive admin action logged in the audit log, since there is only one admin tier.

11. Known limitations and open items

- Non-blocking review means a bad registration has real access to a family's location and fall history until admin catches it. How long that window stays open depends on how often admin checks the review queue.
- Push notification delivery depends on Firebase Cloud Messaging being configured for the web dashboard and any future app.
- The delivery channel for caregiver approval requests still needs a decision.
- The fall incident count does not automatically decrease if an incident later gets marked a false alarm during review. If that matters, the count may need a manual adjustment option for admin or caregivers.